

Yard Sale Market Model: A simplified model for economic inequality

Leonardo Costa Lesage and Ruth Mora Soto
Universitat Politècnica de Catalunya

Can the vast wealth gaps of modern economies emerge from the same statistical laws that govern physical systems? Building on the Yard-Sale model’s random-valuation error mechanism, we design a stylised two-asset market—cash and stock—with a dividend-paying firm, a limit-order book, and agents who differ only in their saving habits. In this *YS-Market* setting, a flat wealth tax χ plus mild valuation noise spans the entire inequality spectrum: $\chi = 0$ drives complete wealth condensation; intermediate χ produces partial oligarchy; and larger χ stabilises a mobile society with a Pareto-tailed distribution. Despite omitting credit, growth, learning, and multi-firm competition, the model shows how a handful of physics-inspired ingredients—bounded stakes, stochastic exchange, and minimal microstructure—reproduce key distributive tensions and provide a sandbox for testing remedies or diagnosing the roots of economic inequality. Simulation code is publicly available at: github.com/ruthilberry/random-asset-exchange.

I. INTRODUCTION AND BACKGROUND

Economic inequality is as old as organised society, yet its persistence—and recent resurgence—in many economies have placed it at the centre of both academic and policy debates. Widening wealth gaps correlate with social unrest, slower growth, and weaker democratic institutions [1, 2]. Therefore, quantifying where inequality comes from—and what mechanisms sustain it—remains a first-order scientific question.

Classical political economy already recognises several broad drivers of unequal outcomes: (i) Productivity-based or “meritocratic” mechanisms. Differences in human capital and the ability to generate high marginal-productivity income can, in principle, justify earnings dispersion (as studied by Tinbergen) [3]. (ii) Intergenerational transmission of wealth. In low-growth environments inherited assets dominate lifetime resources, leading to dynastic wealth accumulation (as popularized by Piketty) [2]. (iii) Financial-market dynamics. When the private return on capital r systematically exceeds the macro growth rate g , wealth concentrates [2]. This means accumulated capital grows faster than labour income; so those who already own capital become ever richer while workers cannot keep pace. Although each mechanism offers valuable qualitative insight, none by itself reproduces the heavy-tailed wealth distributions observed empirically.

Standard growth and business-cycle frameworks (e.g. Solow, Ramsey–Cass, or real-business-cycle models) achieve analytic elegance by collapsing the population into a single representative household. Such models predict average income paths but are silent on the distribution of wealth; they therefore cannot endogenously generate inequality dynamics [4]. Extensions that introduce heterogeneity typically rely on restrictive assumptions (perfect markets, rational expectations, no credit limits) that dampen, rather than amplify, dispersion. This gap has motivated a parallel econophysics literature that borrows tools from statistical physics - and often implements them through large-scale agent-based models - to study wealth exchange and heavy-tailed distributions [5].

Agent based models differ in that they consider an economy made of many different individual agents. A particular class of them, random-asset-exchange models (RAEMs) [6] treat the economy as a large ensemble of agents that stochastically exchange small stakes of wealth [7, 8]. The simplest of these, the Yard-Sale Model (YS), constrains the stake to the poorer agent’s holdings and produces unbiased exchanges with zero expected value. Paradoxically, repeated play drives the system into wealth condensation: a single agent ultimately holds (almost) all wealth [9]. While unrealistic, this pathological limit highlights how microscopic symmetry does not guarantee long term equality. Several extensions have been developed—taxation [10], wealth attained advantage [11], networks [12]. In particular, the addition of taxation regularises the distribution. The further addition of wealth attained advantage and debt constitutes the so-called affine wealth model (AWM). Indeed, this model has achieved Lorenz curves with excellent fit to U.S. and France wealth data [13] as well as realistic Gini coefficients.

Despite their insight and results, most RAEMs only track wealth assuming that assets have an objective value. They also abstract away the institutional details of markets or microeconomic structure. In particular, exchanges occur pairwise and randomly without tracking any individual assets other than total wealth. Our goal is to bridge this gap by embedding the YS’s core idea of judgement mistakes into a financial setting.

We develop a stylized dynamical toy model called the Yard Sale Market model. Specifically, we:

1. Construct a two-time market in which agents hold both cash and a dividend-paying asset traded through a limit-order book.
2. Introduce—(a) a flat wealth tax with lump-sum redistribution and (b) stochastic “valuation errors” that capture bounded rationality and partial information
3. Show that this YS-Market model shows a range of possible magnitudes of inequality, from total oligarchy (no taxes) to Pareto tail distribution (moderate taxes).

Because the framework remains conceptually simple, it offers a starting point for testing alternative hy-

potheses—inheritance, growth shocks, progressive taxation—within a stylized economic model.

Our contribution is therefore two-fold: (i) we propose a minimal set of micro-structural rules that transforms the Yard-Sale mechanism into a plausible market economy, and (ii) we demonstrate that basic institutional features—cash and a dividend-paying stock, flat wealth levies, and human error—already produce realistic, heavy-tailed wealth distributions.

The remainder of the paper is organised as follows. Section 2 reviews the results from classical Yard-Sale (YS) benchmark. Section 3 introduces the YS-Market model and Section 4 discusses the numerical results. Finally, Section 5 reflects on the remaining open avenues for empirically testing inequality-generating mechanisms building on the YS-Market model.

II. YARD SALE MODEL

The classical Yard-Sale Model (YS) [7, 14] captures stochastic wealth exchange with a single rule: whenever two agents i and j meet, they trade a stake $\Delta w = \varepsilon \min(w_i, w_j)$. Despite the locally fair trade, repeated play drives the system toward wealth condensation. This exchange rule captures the dynamics of informal markets, such as yard sales, where transactions often occur at prices that slightly deviate from the intrinsic value of goods due to limited information and bounded rationality among agents. It is precisely these noisy valuations or disagreements that motivate the trades.

To quantify how far the model departs from equality we track two standard indicators:

a. Lorenz curve $L(p)$ plots the cumulative share of total wealth held by the bottom p fraction of the population, when individuals are ranked from poorest to richest. If everyone had exactly the same wealth, the curve would follow the diagonal $L(p) = p$. Deviations below this line reflect inequality.

b. Gini coefficient Measures the area between the Lorenz curve and the line of perfect equality, with $G = 0$ meaning complete equality and $G = 1$ meaning all wealth is held by a single agent.

We reproduce the well-known result that the baseline Yard-Sale Model leads to complete wealth condensation in the absence of redistribution mechanisms [5, 7, 14], as shown in Fig. 1. This outcome is clearly visible in the Lorenz curve bending strongly away from the diagonal. Extensions such as wealth-attained advantage (WAA), favoring richer agents in trades, [10] and redistributive taxation [15] modulate this behaviour, respectively regularizing or amplifying inequality, as illustrated in Fig.1.

III. YS MARKET MODEL

The Yard-Sale framework of Sec. II presented inequality generated from trades where agents make a mistake in

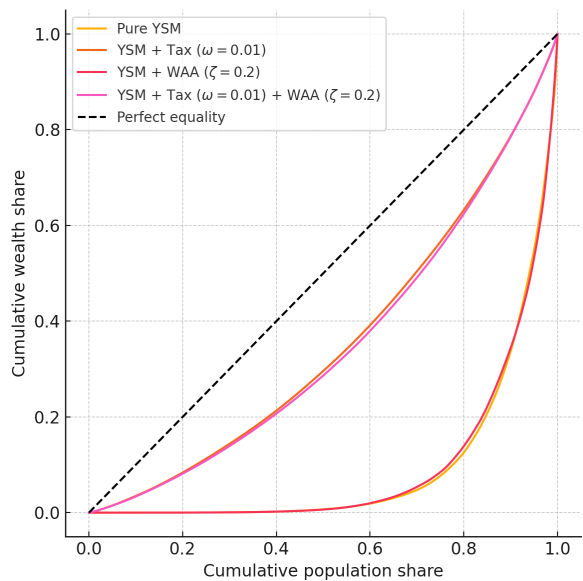


FIG. 1: Lorenz curves for the Yard-Sale Model (YS) with extensions. All curves correspond to steady-state distributions.

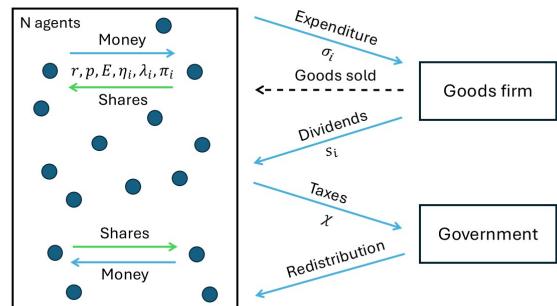


FIG. 2: Key behaviour of the model. Goods sold are not tracked

estimating the value of some good. This basic precept of mistakes in judgement driving exchanges is explored inside a very simple financial market grounded on an even simpler goods market.

We call this model the YS Market model. It includes N agents (individuals), one firm and a government. We consider a market of firm shares that are valued for the dividends they return. These dividends come from the firm's profit. Agents make judgement mistakes at random when estimating the value of future dividends and these differences in judgement drive a share market implemented with an order book. Finally, a government applies taxes and equally distributes them. A sketch of the model behaviour is represented in 2.

In what follows we describe the behaviour in more detail. There are N different agents. Their parameters will be denoted with an i subscript ($i = 1, 2, \dots, N$) and can

be all different. At each time step, agents have some non-negative real amount of money m_i and some non-negative integer amount of shares from a firm s_i . They also have some parameters related to consumption (σ_i) and market behaviour (λ_i, η_i, π_i). Finally, there are some global parameters related to value estimation (r), judgement errors (p, E) and taxation (χ).

The state of the system evolves in discrete time steps. Every time step consists of four phases: (1) Agents buy goods from the firm, (2) Firm revenue is distributed among agents through dividends, (3) Agents buy and sell shares amongst each other, (4) The government applies taxes on money and redistributes collected money.

Goods market. Agents are assumed to have a desire to consume goods, produced by a single firm. The amount of goods produced is exactly that demanded and all goods produced are consumed in the same time step. Production is also abstracted, so we assume production costs are zero and thus all profit is revenue.

The desire to save versus consume is encoded by a savings parameter $0 \leq \sigma_i \leq 1$. At each time step, an agent with m_i money will save $\sigma_i m_i$ and spend $(1 - \sigma_i)m_i$ on goods. This behaviour actually corresponds to maximizing a particular utility function depending on consumption and saved money.

Dividends. All company revenue is profit and is distributed in the same time step among agents proportionally to the amount of shares s_i they hold.

Share market. All agents value shares by the time-discounted total money they will receive from them through dividends. In particular, 1 unit of money t time-steps from now is valued as $(1 - r)^t$ units of money in the future. This encodes the preference for money now with respect to money in the future that agents have. For the sake of simplicity, r is assumed to be the same for all agents. Taking d_0 , the dividend return in this time-step as the best estimation for future dividend returns, a simple calculation yields that the estimated time-discounted value of future dividend returns is d_0/r . We call this the "rational estimation": $RE = d_0/r$.

If everyone agreed on the estimation of the value of a share, no trades would ever happen. We assume that agents estimate the value of a share as $RE \cdot J_i$, where J_i is a judgement factor. It will be 1 with probability $1 - p$ and follow a $N(1, E^2)$ distribution with probability p . The normal distribution is clipped to $[0.1, 3]$ to avoid very extreme errors. Again, for the sake of simplicity, E, p are assumed to be the same for all agents.

Agents have some edge requirement, meaning they will only buy or sell if they believe they are winning at least a certain fraction (η_i) of the exchange value by performing the trade. To avoid extreme daily trading behaviour, agents are assumed never spend all their money on stocks or sell all their stocks in a single time step. In a particular, they never buy or sell for a total value larger than a limit fraction (λ_i) of their total self-estimated wealth (money plus estimated value of shares held). Addition-

ally, a prudence parameter (π_i) is added, such that agents will never sell shares if they only have a fraction π_i of their initial shares left.

Agents desires to buy/sell are stored in an order book that is reset at each time step. Agents have the option to make a to buy and or sell at the best price available in the market; or, in case the available prices do not satisfy their edge requirement, place a bid or ask. Agents enter the market at a random order that is different every day.

Taxes. At the end of the time step, the government collects a fraction (χ) of the wealth of each individual and redistributes all collected money uniformly among agents (each gets $1/N$ of the collected money).

IV. ANALYSIS AND EXPERIMENTS

We simulate the YS-Market model in Python/NumPy and summarise the key results below.

We begin by examining long term behaviour omitting the trade stage and the taxes phase. Thus the system becomes a deterministic dynamical system. It can be easily justified (supposing $\sigma_i > 0 \forall i$) that the system will always converge to an equilibrium irrespective of initial conditions. On the long term, agents spend as much as they earn through dividends, so we get:

$$\text{Consumption spending} = m_i(1 - \sigma_i) = \text{Earnings} \propto s_i$$

This has the implication that in the long term, consumption will only depend on shares held, while total amount of money held will depend both on the amount of shares and the savings parameter. Agents that save more will have more savings, but will consume the same in the long term. This means that there can be a relevant difference between wealth inequality and consumption inequality, and what is more directly linked to welfare is consumption inequality, but we have not studied further in this work.

Now we will focus on the effects of introducing the share market with error judgements. For the sake of simplicity, we will assume that all agents have the same parameters. Given that agents will be able to exchange shares for money, neither s_i nor m_i would be appropriate assessments of an agent wealth. We will instead look at the distribution of total wealth $m_i + RE \cdot s_i$ (as estimated by a rational observer).

Without taxes or prudence, we observe that in the long term, all agents except for one lose all their shares. To graphically represent the wealth distribution, we plot the Lorenz curve (3). The Gini coefficient evolution in time is also plotted.

Maximum inequality is observed, and further investigation shows that all stocks are owned by a single agent. This means that agents earn no income, so once they become poor enough they are completely unable to buy any shares back, they will never earn income again and will stay in a perfect poverty trap. This behaviour is analogous to the yard sale model wealth condensation with no

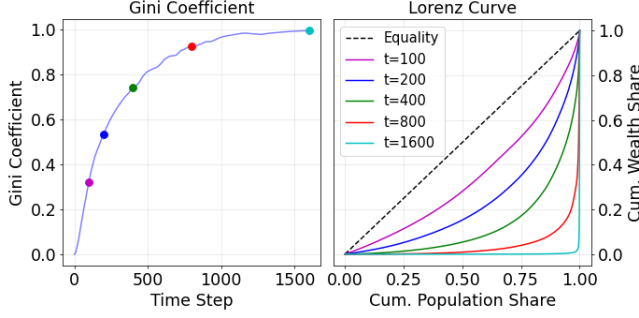


FIG. 3: $N = 1000$ agents simulation with no taxes ($\chi = 0$) or prudence ($\pi_i = 0$). All agents start with $m_i = 1, s_i = 1000$. All agents have the same parameters: $\sigma_i = 0.99, \eta_i = 0.01, \lambda_i = 0.20, r = 0.1$. Error parameters are $E = 0.2, p = 0.01$. Gini coefficient time evolution and the Lorenz curves at some representative times are shown. Inequality increases over time towards a state of maximum inequality.

redistribution. We have thus illustrated a general principle that whenever it there is a positive probability of falling into a poverty trap, after sufficient time agents will fall. And if the poverty trap is unescapable, all agents will eventually fall and stay there (except for the richest agent).

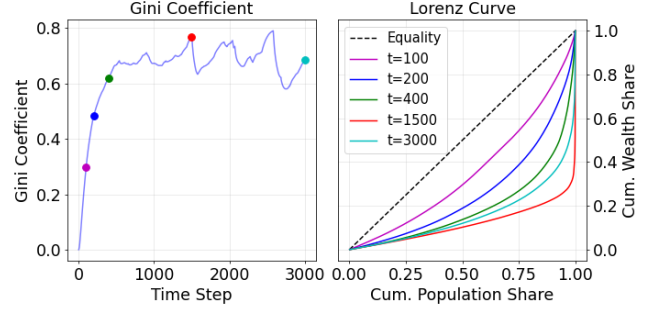
We finally look at how adding the taxes stage or prudence stabilizes the distribution (see 4). With either of them we avoid maximum inequality. With taxation, agents that have lost almost all wealth will recover some, and for large enough values of taxation, will be able to re-enter the share market. With prudence, agents will never fall into a poverty trap, as they never sell below a threshold of shares.

When varying the values of χ and π_i we observe, as expected, that higher values correspond to lower inequality. It is interesting to note that we do not reach a steady state. Random oscillations are considerable (up to 0.2 changes in Gini in our examples). This effect would probably be much lower for larger values of N , although we have not tested this thoroughly. In fact, in the case of the extended yard sale model, it is well-established that they reach a steady state (in the continuous limit $N \rightarrow \infty$) [10]. Another interesting feature is that we sometimes observe a "partial oligarchy" (as it is called in the context of the EYSM). That is, we observe that sometimes a few agents (as few as one) hold all wealth. It would be interesting to investigate if this also happens in the continuous limit, as is the case for the EYSM.

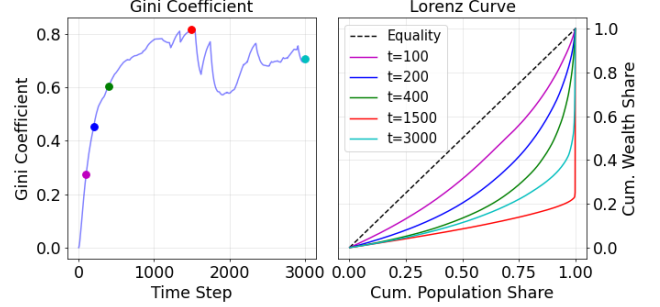
V. OUTLOOK

The main next step would be to expand the model to become a more interesting toy model, allowing one to

compare the inherited wealth and $r > g$ hypothesis of wealth. A first step would be to give agents a lifetime, such that agents behave differently as young in a fraction



(a) Only taxation. $\chi = 0.0015, \pi_i = 0.00$



(b) Only prudence. $\chi = 0.0000, \pi_i = 0.15$

FIG. 4: $N = 1000$ agents. All agents start with $m_i = 1, s_i = 1000$ and have the same parameters $\sigma_i = 0.99, \eta_i = 0.01, \lambda_i = 0.20$. The other parameters are $r = 0.1, E = 0.2, p = 0.01$. Gini coefficient time evolution and Lorenz curves at some representative times are shown. Taxation and prudence values are chosen to get similar values of the Gini coefficient.

of their lifetime and as old in the remainder, with different consumption trends. When an agent dies, it would leave its a wealth to a new agent (its offspring). On the other hand, to explore the $r > g$ hypothesis, we would have to move our model closer to a minimal model of the economy [16], adding a capital market, a labour market, etc.

Another way in which this work could be continued is to connect our posited random valuation errors to a differing knowledge of random demand shocks, as are considered in mainstream models.

Additionally, it would be fruitful to more carefully examine the behaviour of agents by altering the error probabilities between agents to more strongly establish the relationship between judgement errors and poverty. It would also be interesting in this direction to look at the results of making poor agents less likely to commit errors, as research shows that lower-income groups tend to be more careful in their financial decisions [17].

-
- [1] D. Acemoglu and J. A. Robinson, *Why Nations Fail: The Origins of Power, Prosperity, and Poverty* (Crown Publishing Group, 2012).
 - [2] T. Piketty, *Capital in the Twenty-First Century* (Harvard University Press, 2014).
 - [3] J. Tinbergen, Income distribution: Analysis and policies, in *Income Distribution*, edited by J. Ekblom (North-Holland, 1975) pp. 27–32.
 - [4] R. M. Solow, A contribution to the theory of economic growth, *Quarterly Journal of Economics* **70**, 65 (1956).
 - [5] A. Dragulescu and V. M. Yakovenko, Statistical mechanics of money, *The European Physical Journal B* **17**, 723 (2000).
 - [6] G. M., Twenty-five years of random asset exchange modeling, *The European Physical Journal B* **97**, 69 (2024).
 - [7] J. Angle, The surplus theory of social stratification and the size distribution of personal wealth, *Social Forces* **65**, 293 (1986).
 - [8] A. Chakraborti, Statistical mechanics of money: How saving propensity affects its distribution, *The European Physical Journal B* **17**, 167 (2000).
 - [9] J. M. . M. J. Boghosian, B.M., An h theorem for boltzmann’s equation for the yard-sale model of asset exchange, *Journal of Statistical Physics* **161**, 1331 (2015).
 - [10] B. M. Boghosian, Kinetics of wealth and the pareto law, *Phys. Rev. E* **89**, 042804 (2014).
 - [11] B. M. Boghosian, A. Devitt-Lee, M. Johnson, J. Li, J. A. Marcq, and H. Wang, Oligarchy as a phase transition: The effect of wealth-attained advantage in a fokker–planck description of asset exchange, *Physica A: Statistical Mechanics and its Applications* **476**, 15 (2017).
 - [12] R. Bustos-Guajardo and C. F. Moukarzel, Yard-sale exchange on networks: wealth sharing and wealth appropriation, *Journal of Statistical Mechanics: Theory and Experiment* **2012**, P12009 (2012).
 - [13] Boghosian, The affine wealth model: An agent-based model of asset exchange that allows for negative-wealth agents and its empirical validation, *Physica A: Statistical Mechanics and its Applications* **516**, 423 (2019).
 - [14] A. Chakraborti, Statistical mechanics of money: How saving propensity affects its distribution, *The European Physical Journal B* **17**, 167 (2002).
 - [15] B. M. Boghosian, A. Devitt-Lee, and M. Johnson, Wealth inequality and slider–bar taxation in the yard-sale model, *Physical Review E* **95**, 042125 (2017).
 - [16] C. K. Chan and K. Steiglitz, An agent-based model of a minimal economy, Dept. of Computer Science, Princeton University Princeton (2008).
 - [17] S. Mullainathan and E. Shafir, *Scarcity: Why having too little means so much* (Macmillan, 2013).